

Static Correlation Functions in **nssDMRG**

Mathematical Conventions and Implementation Notes

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Contents

1 Purpose and scope

The implementation in `DMRG_MEASURE` explicitly constructs the operator product $O_A(i)O_B(j)$ in the final renormalized basis and evaluates its expectation value on the stored superblock state. The principal difficulties are that the two operators may belong to the same DMRG block or to opposite blocks, and that fermionically odd operators require the correct parity string.

The geometrical cases are

$$i, j \in L, \quad i, j \in R, \quad i \in L, j \in R, \quad (1)$$

plus the special local case $i = j$.

2 Correlation functions and public interface

The generic routine evaluates

$$C_{AB}(i, j) = \langle \Psi | O_A(i) O_B(j) | \Psi \rangle, \quad (2)$$

where $|\Psi\rangle$ is the stored superblock state in the target quantum-number sector. If `connected=.true.`, the result is instead

$$C_{AB}^c(i, j) = \langle O_A(i) O_B(j) \rangle - \langle O_A(i) \rangle \langle O_B(j) \rangle. \quad (3)$$

The preferred key-based interface is

```
corr = Measure_Corr_DMRG(keyA, keyB, posA, posB)
corr = Measure_Corr_DMRG(keyA, keyB, posA, posB, connected=.true.)
```

In this form, the matrix, quantum-number shift, and operator type are obtained together from `LIST_OPERATORS`.

The lower-level interface is intended for conjugated or externally composed operators:

```
corr = Measure_Corr_DMRG(OpA, dqA, OpB, dqB, posA, posB, typeA, typeB)
corr = Measure_Corr_DMRG(OpA, dqA, OpB, dqB, posA, posB, typeA, typeB, &
    connected=.true.)
```

Here the caller is responsible for supplying consistent shifts and operator types.

3 Quantum-number shifts

For every operator the convention is

$$dq(O) = Q_{\text{out}} - Q_{\text{in}}, \quad (4)$$

or, equivalently,

$$O|q\rangle \mapsto |q + dq(O)\rangle. \quad (5)$$

For a product of Abelian operators,

$$dq(O_A O_B) = dq_A + dq_B. \quad (6)$$

Since the expectation value is evaluated within a fixed total sector,

$$\langle O_A O_B \rangle \neq 0 \implies dq_A + dq_B = 0. \quad (7)$$

Failure of this condition is a selection rule rather than an invalid input; the routine therefore returns zero.

For example,

$$dq(S^+) + dq(S^-) = 0, \quad (8)$$

$$dq(c_{\uparrow}^{\dagger}) + dq(c_{\uparrow}) = (+1, 0) + (-1, 0) = (0, 0). \quad (9)$$

Conversely, in a particle-number eigenstate,

$$\langle c_{i\uparrow}^{\dagger} c_{j\downarrow}^{\dagger} \rangle = 0, \quad (10)$$

because its total shift is $(+1, +1)$. Such an operator may still represent a transition between different sectors; the present routine computes an expectation value within one fixed sector.

4 Fermionic grading

The quantum-number shift does not determine the fermionic sign structure. One must also specify the $\mathbb{Z}_2$ grading

$$p(O) = N_f(O) \bmod 2, \quad (11)$$

where $N_f(O)$ is the number of elementary c or $c^{\dagger}$ factors in O . Examples are

$$p(c) = p(c^{\dagger}) = 1, \quad (12)$$

$$p(cn) = 1, \quad (13)$$

$$p(n) = p(c^{\dagger}c) = 0, \quad (14)$$

$$p(S^+) = p(c_{\uparrow}^{\dagger}c_{\downarrow}) = 0, \quad (15)$$

$$p(\Delta) = p(c_{\downarrow}c_{\uparrow}) = 0, \quad (16)$$

$$p(P) = p((-1)^N) = 0. \quad (17)$$

Composition obeys

$$p(O_A O_B) = p(O_A) + p(O_B) \pmod{2}. \quad (18)$$

In the current implementation, `type="fermionic"` denotes an odd operator. For expectation values in a state of definite fermion parity, the total operator must be even. Thus even–even and odd–odd products are allowed; even–odd products vanish. An odd–odd product is even overall, but requires a parity string.

The shift and grading are independent metadata. For example, the pair operator $\Delta^{\dagger} = c_{\uparrow}^{\dagger}c_{\downarrow}^{\dagger}$ is even but has a nonzero quantum-number shift.

5 Superblock state and sector decomposition

The stored state is expanded in the product basis of the final left and right blocks:

$$|\Psi\rangle = \sum_{\alpha,\beta} \Psi_{\alpha\beta} |\alpha\rangle_L \otimes |\beta\rangle_R. \quad (19)$$

If $|\alpha\rangle_L \in \mathcal{H}_L(q)$ and the total quantum number is Q , the compatible right state belongs to $\mathcal{H}_R(Q - q)$. Consequently,

$$|\Psi\rangle = \bigoplus_q |\Psi_q\rangle, \quad |\Psi_q\rangle \in \mathcal{H}_L(q) \otimes \mathcal{H}_R(Q - q). \quad (20)$$

During measurement initialization, LI and RI store the block states associated with each superblock sector. The inverse maps **Lmap** and **Rmap** convert a full block-state index into its index inside a sector. They are used to construct rectangular operator blocks efficiently when $dq \neq 0$.

6 Local case: equal positions

For $i = j$, the product is first formed in the local Hilbert space, preserving the requested order:

$$O_{AB}(i) = O_A(i)O_B(i). \quad (21)$$

It is then embedded into the growing block and transported to the final renormalized basis. No Jordan–Wigner string is introduced in this case. The ordering is essential; for example,

$$c_i^\dagger c_i = n_i, \quad c_i c_i^\dagger = 1 - n_i. \quad (22)$$

7 Transport through block growth

Let U_n map the enlarged block basis at step n to the retained basis. An operator transforms as

$$\tilde{O}^{(n)} = U_n^\dagger O^{(n)} U_n. \quad (23)$$

After the next site is introduced,

$$O^{(n+1)} = \tilde{O}^{(n)} \otimes X_{n+1}, \quad (24)$$

where X_{n+1} is either a requested local factor or the identity. In schematic form,

$$O^{(n)} \longrightarrow U_n^\dagger O^{(n)} U_n \longrightarrow \left(U_n^\dagger O^{(n)} U_n \right) \otimes X_{n+1}. \quad (25)$$

The actual tensor-product ordering is chosen consistently with left/right block growth and with the OBC/PBC convention.

The generic constructor **Build_Product_Block_DMRG** performs this operation. It maps physical positions to growth indices using **b2gMap**, starts with the first nontrivial factor entering the block, inserts identities at all unrelated growth steps, and continues until the final block basis is reached.

8 Why same-block products must be composed before truncation

Suppose that both operators belong to the same block. The correct representation of their product in the retained basis is

$$U^\dagger O_A O_B U. \quad (26)$$

Projecting the operators separately and multiplying afterwards gives instead

$$(U^\dagger O_A U)(U^\dagger O_B U) = U^\dagger O_A \underbrace{U U^\dagger}_{\mathcal{P}} O_B U. \quad (27)$$

Since U is generally rectangular,

$$U U^\dagger = \mathcal{P} \neq I, \quad (28)$$

and therefore

$$U^\dagger O_A O_B U \neq (U^\dagger O_A U)(U^\dagger O_B U) \quad (29)$$

in general. The local factors must consequently be joined before all later truncations.

9 Parity-even operators in the same block

For parity-even operators at distinct positions, `Build_Corr_Block_DMRG` delegates to the generic product constructor. The procedure is:

1. map each physical position to its block-growth index;
2. determine which nontrivial factor enters first;
3. build that first factor;
4. apply each subsequent U_n ;
5. insert identities at intermediate sites;
6. insert the other factor when its site enters;
7. continue to the final block basis.

The resulting operator is either O_{AB}^L or O_{AB}^R , and its average is evaluated as

$$\langle \Psi | O_{AB}^L \otimes I_R | \Psi \rangle \quad \text{or} \quad \langle \Psi | I_L \otimes O_{AB}^R | \Psi \rangle. \quad (30)$$

10 Odd fermionic operators in the same block

For two odd operators at $i < j$, the tensor-product representation used by `nssDMRG` is

$$O_i O_j \longrightarrow (O_i P_i) P_{i+1} \cdots P_{j-1} O_j, \quad P_k = (-1)^{N_k}. \quad (31)$$

For example,

$$c_i^\dagger c_j \longrightarrow (c_i^\dagger P_i) P_{i+1} \cdots P_{j-1} c_j. \quad (32)$$

The factor list passed to the generic constructor is therefore

$$[O_i P_i, P_{i+1}, \dots, P_{j-1}, O_j]. \quad (33)$$

If the requested order has the first argument on the right, the string is still assembled in spatial left-to-right order. Exchanging the two odd endpoints once produces

$$O_A(i) O_B(j) = -O_B(j) O_A(i), \quad i \neq j, \quad (34)$$

and the final operator is multiplied by -1 .

11 Operators on opposite sides of the cut

For $i \in L$ and $j \in R$, parity-even operators give

$$C_{AB}(i, j) = \langle \Psi | O_A^L(i) \otimes O_B^R(j) | \Psi \rangle. \quad (35)$$

The two factors are independently transported to the final bases of their blocks. This expectation value cannot be evaluated by a same-block average, because neither tensor factor is the identity. It requires an explicit superblock contraction.

11.1 Sector transitions

The left operator maps

$$O_L : \mathcal{H}_L(q_{\text{in}}) \longrightarrow \mathcal{H}_L(q_{\text{out}}), \quad q_{\text{out}} = q_{\text{in}} + dq_L. \quad (36)$$

Writing $q = q_{\text{out}}$ gives

$$q_{\text{in}} = q - dq_L. \quad (37)$$

Thus the rows of the filtered operator belong to the output sector q , and its columns belong to the input sector $q - dq_L$. The corresponding rectangular block is

$$A_q = O_L|_{\mathcal{H}_L(q-dq_L) \rightarrow \mathcal{H}_L(q)}. \quad (38)$$

Conservation of the total quantum number requires

$$dq_R = -dq_L, \quad (39)$$

so that the right rectangular block performs the compatible transition. For each output sector q , the contribution is

$$C_q = \langle \Psi_q | A_q \otimes B_q | \Psi_{q-dq_L} \rangle, \quad (40)$$

and the full correlation is

$$C_{AB}(i, j) = \sum_q C_q. \quad (41)$$

11.2 Matrix form of the contraction

For a fixed sector, reshape the superblock vector as a matrix $\Psi_{\alpha\beta}$, where α and β are left- and right-block indices. Then

$$\Phi_{\alpha\beta} = \sum_{\alpha', \beta'} A_{\alpha\alpha'} B_{\beta\beta'} \Psi_{\alpha'\beta'}, \quad (42)$$

or, subject to the vectorization convention,

$$\Phi = A \Psi B^\top. \quad (43)$$

The expectation value is

$$C = \sum_{\alpha, \beta} \Psi_{\alpha\beta}^* \Phi_{\alpha\beta}. \quad (44)$$

The implementation applies the two factors successively and never constructs the full Kronecker matrix $A \otimes B$. This substantially reduces memory usage.

11.3 Distributed contraction

In MPI mode the superblock vector is distributed among processes. The same mathematical operation is performed in two data layouts:

$$\Psi \xrightarrow{B} X \xrightarrow{\text{MPI transpose}} X^\top \xrightarrow{A} \Phi^\top \xrightarrow{\text{MPI transpose}} \Phi. \quad (45)$$

Each process evaluates its local scalar product

$$C_r = \langle \Psi_r | \Phi_r \rangle, \quad (46)$$

and the total value is obtained through

$$C = \sum_r C_r. \quad (47)$$

12 Odd fermionic string across the cut

For $i \in L$, $j \in R$, and $i < j$,

$$O_i O_j = (O_i P_i) P_{i+1} \cdots P_L P_{L+1} \cdots P_{j-1} O_j. \quad (48)$$

The string is split into

$$O_L^{\text{string}} = (O_i P_i) P_{i+1} \cdots P_L, \quad (49)$$

$$O_R^{\text{string}} = P_{L+1} \cdots P_{j-1} O_j. \quad (50)$$

The correlator becomes

$$C_{AB}(i, j) = \langle \Psi | O_L^{\text{string}} \otimes O_R^{\text{string}} | \Psi \rangle. \quad (51)$$

For adjacent sites at the cut, this reduces to the convention already used for hopping terms,

$$(c_L^\dagger P_L) \otimes c_R. \quad (52)$$

If the requested operator order is right-to-left, one exchange of odd endpoints produces the additional minus sign.

13 Connected correlations

For `connected=.true.`, the disconnected contribution is subtracted after the full two-point function has been computed. A one-point average can be nonzero only for an even operator with zero shift:

$$dq(O) = 0, \quad p(O) = 0. \quad (53)$$

If either condition fails, the one-point function vanishes by symmetry, and the implementation avoids evaluating it.

14 Spin-spin wrapper

The spin wrapper uses

$$\mathbf{S}_i \cdot \mathbf{S}_j = S_i^z S_j^z + \frac{1}{2} S_i^+ S_j^- + \frac{1}{2} S_i^- S_j^+. \quad (54)$$

Accordingly,

$$\langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle = \langle S_i^z S_j^z \rangle + \frac{1}{2} \langle S_i^+ S_j^- \rangle + \frac{1}{2} \langle S_i^- S_j^+ \rangle. \quad (55)$$

Each term separately satisfies $dq_A + dq_B = 0$. Spin operators are fermionically even, including the electronic realization $S^+ = c_\uparrow^\dagger c_\downarrow$, and therefore require no parity string.

15 Spin-orbital-resolved density correlations

The density wrapper returns the matrix

$$C_{ab}(i, j) = \langle n_{ia} n_{jb} \rangle, \quad a, b = 1, \dots, N_{so}, \quad (56)$$

where

$$N_{so} = N_{\text{spin}} N_{\text{orb}}, \quad a = i_{\text{orb}} + (i_{\text{spin}} - 1) N_{\text{orb}}. \quad (57)$$

For each spin-orbital,

$$n_a = c_a^\dagger c_a, \quad dq(n_a) = 0, \quad p(n_a) = 0. \quad (58)$$

Consequently, no fermionic string is needed. In connected mode,

$$C_{ab}^c(i, j) = \langle n_{ia} n_{jb} \rangle - \langle n_{ia} \rangle \langle n_{jb} \rangle. \quad (59)$$

The total-density correlation can be recovered as

$$\langle N_i N_j \rangle = \sum_{a,b=1}^{N_{so}} C_{ab}(i, j), \quad N_i = \sum_a n_{ia}. \quad (60)$$

16 Complete measurement flow

The generic measurement proceeds as follows:

1. Initialize the measurement state and load the blocks, rotation matrices, sector decomposition, and superblock vector.
2. Validate positions and the dimensions of the two shifts.
3. Apply the selection rules

$$dq_A + dq_B = 0, \quad p_A + p_B = 0 \pmod{2}.$$

4. If $i = j$, compose the two operators locally in the requested order.
5. If both positions belong to one block, build the complete product before subsequent truncations. Insert the parity string for two odd endpoints.
6. If the positions belong to opposite blocks, build the two final block operators separately, split the fermionic string at the cut when necessary, filter all rectangular sector transitions, and evaluate $\langle \Psi | O_L \otimes O_R | \Psi \rangle$.
7. If requested, subtract the disconnected one-point contribution.

The two independent pieces of physical metadata are therefore

$$\boxed{dq(O)} \quad \text{and} \quad \boxed{p(O)}. \quad (61)$$